Samuel Levens

Senior Software Engineer · Generative AI products

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Full-stack TypeScript engineer and tech lead across startups, Atlassian and Relume. I build generative AI products and the tooling that lets AI agents work inside real systems: I led Relume Studio, an AI image generation product, from prototype to launch, and wrote the MCP server that lets Relume's AI build with its component library. I take new product areas to production and harden them with tests, metrics and explicit failure handling.

EXPERIENCE

Senior Software Engineer · Relume

Jan 2026 – present

Tech lead for two product areas on Relume's Nova platform: Relume Studio, a generative AI image product, and the component Builder that produces Relume's component library and every published site. Merged 691 pull requests and reviewed 275 in the first ten months.

Relume Studio: generative AI image product tech lead · Sep 2026 –

- Rebuilt Studio from a prototype into a production application in three weeks, from the first foundation pull request to internal release, and led it through public launch. Authored 74% of Studio's pull requests.
- Led delivery of the core generative features: Styles generated from reference images, People and Scenes for consistent characters and environments across generations, image refinement, and a Library synced into the site editor.
- Built the generation pipeline on BullMQ with fallback across OpenAI and Google image models, rate-limit retries, per-user and fleet-wide queue limits, and hand-back of in-flight jobs on deploy.
- Metered every generation per token against per-plan spend caps. Instrumented the product before launch with Prometheus and Grafana timings, per-job cost in PostHog, Sentry, failure-rate alerts and a kill switch for all paid work.
- Designed AI image upscaling on a Cloudflare Worker at $0.0005 per image, against $0.004 to $0.17 for the alternatives evaluated. Mapped the model's undocumented input limits through three spikes. The editor team has adopted the Worker.
- Ran a 10-area launch readiness review that raised 51 issues; 39 were fixed and 11 partly fixed within one day, and the go/no-go review returned GO. Changed AI quality and safety checks from blocking errors to recorded warnings after false positives failed paid generations.

AI agents and the component Builder tech lead · Feb – Sep 2026

- Wrote the Builder MCP server, which lets Relume's internal AI and designers using Claude Code operate the component library builder directly. It replaced the Builder's in-app chat. Hardened it for production with secure pairing, reconnection backoff and port-conflict recovery.
- Built a `qa_component` check that validates a component against the schema, the editor and published output and returns a machine-readable result, and added selection criteria that agents use to choose components.
- Designed the component model: variants, slots, sibling component families, conditional properties and per-instance overrides. Led the schema rework across 270 files, with all 1,989 tests passing.
- Added Git-style branching to the component library, and concurrent-edit protection with three-way merges so designers and AI agents can edit simultaneously without overwriting each other.
- Moved published sites from React hydration (230 KB) to server-rendered HTML with a 10 KB vanilla runtime, and unified the render pipelines, removing the legacy HTML generator.
- Halved backend idle memory after a heap-snapshot investigation, eliminating out-of-memory crashes in preview pods without raising limits. Introduced Sentry error tracking and OpenTelemetry distributed tracing.

Software Engineer · Atlassian, Commerce — Jan 2022 – 2025

- Built and maintained the billing console UI and its backend-for-frontend with React, Node.js and GraphQL.
- Led the migration of the billing experience into Jira and Confluence using a micro-frontend architecture.
- Upgraded the codebase from React 17 to 18, resolving compatibility issues and optimising Suspense usage.
- Rebuilt the ESLint and pre-commit pipeline to reduce noise and CI failures.
- Contributed to architecture decisions for the billing platform to enable code sharing across the Commerce department.

Senior Software Engineer · FrankieOne — Sep 2021 – Jan 2022

- Rewrote the identity verification flow in React, replacing a legacy Vue implementation, to support client customisation and mobile use.
- Designed a JavaScript SDK for embedding KYC and AML verification into client web and mobile apps.

Systems Architect and Full Stack Developer · GADA Technology (Internmatch) — Jun 2020 – Jan 2022

- Promoted from a front end contract role to systems architect.
- Rebuilt the React front end for rapid delivery on large B2B integrations, with client-specific feature toggles.
- Introduced CI/CD and automated testing with GitHub Actions and Cypress, and established accessibility standards.

Front End Developer · Eventhub — Jan 2019 – Jun 2020

- Built React components and ticket purchasing flows on a greenfield product, and improved accessibility and responsiveness.

Business development · Groupon, NSW Business Chamber, Specifier — 2015 – 2018

Sales, client onboarding and stakeholder outreach.

SKILLS

Generative AI	OpenAI and Google image models, Vercel AI SDK, Mastra, Model Context Protocol servers, agent validation checks
AI-native workflow	Claude Code and Codex agents in parallel git worktrees, plan-first delivery, agent-run code reviews, agent guidance files
Frontend	TypeScript, React, MobX, TanStack Query, React Router, micro-frontends, accessibility
Backend	Node, Hono, GraphQL, Prisma, PostgreSQL, BullMQ, Redis
Edge and cloud	Cloudflare Workers, Pages, Images and R2, AWS, Kubernetes, OpenTofu, Docker
Quality and ops	Vitest, Jest, Playwright, Cypress, Sentry, OpenTelemetry, Prometheus, Grafana, PostHog, Buildkite, GitHub Actions

SIDE PROJECTS

Mobile games in Unity: The Patient Machine, Dicey Defense and Rogue Party, built on pure C# cores for iOS and Android.

agent-dispatch: local coordinators for Claude and Codex workers, with a terminal-style control console.

EDUCATION AND COMMUNITY

University of Sydney · 2016
Bachelor of Engineering (Mechatronics) and Bachelor of Commerce (International Business)

Career mentor, headspace · 2023 – 2024
Volunteer career mentor for young people.